

Intro to Deep Learning

COP 4613 - Machine Learning Concepts Part 1

Module 3 lecture 1

• i Deeplearn 2023

module 2

p1 Deep Learning/Classification Concepts

Lecture 2 module 3

- Face recognition is a supervised model because it has an image of you to confirm recognition
- Speech recognition also supervised
- Bioinformatics - identifying genes responsible for a specific disease

- unsupervised

• Medical diagnosis

- supervised

• clustering last segmentation

- segment lungs from other part of body
pixels that are similar get clustered
puts structures together

~~COP 4613~~ MAS 2103 L3 LATE & review

1.2.00

augmented matrix:

$$\left[\begin{array}{ccc|c} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{array} \right] \quad \neq 2$$

reduce to RREF

$$\begin{array}{l} R_2 \leftrightarrow R_1 \\ R_3 + 2R_1 \end{array} \left[\begin{array}{ccc|c} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & -4/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$